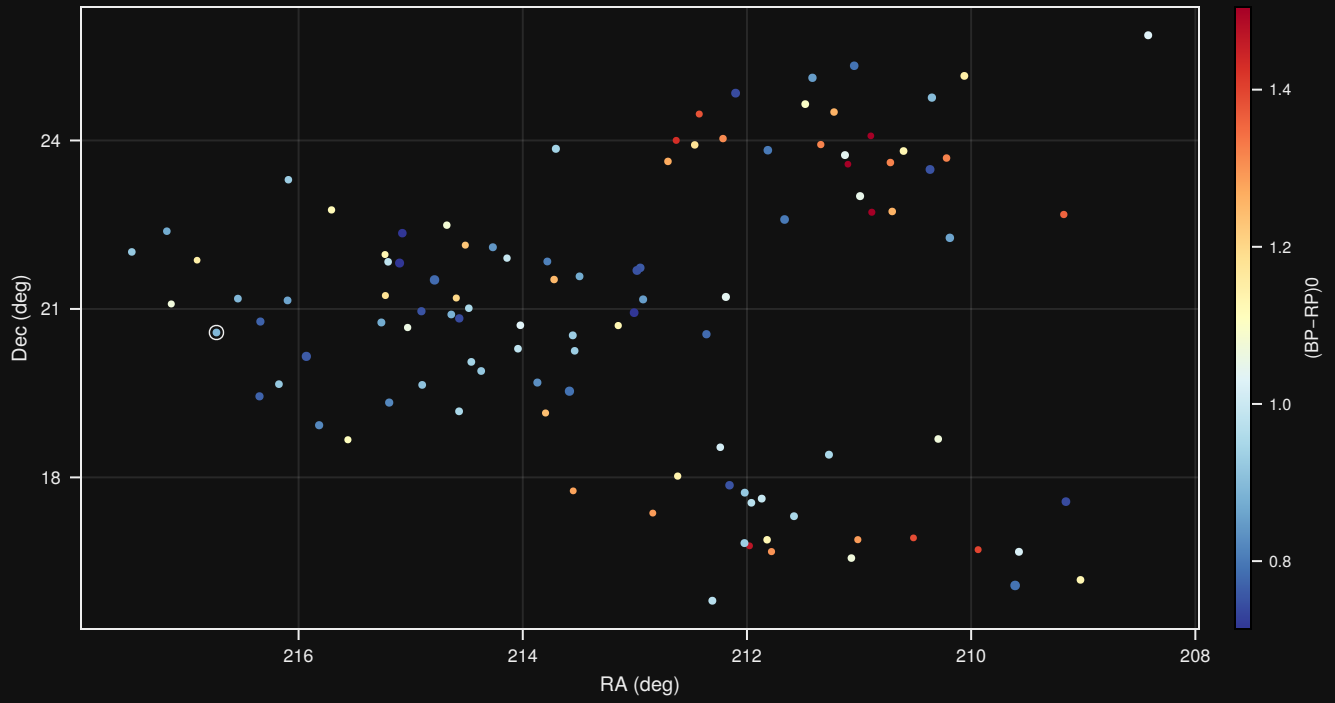
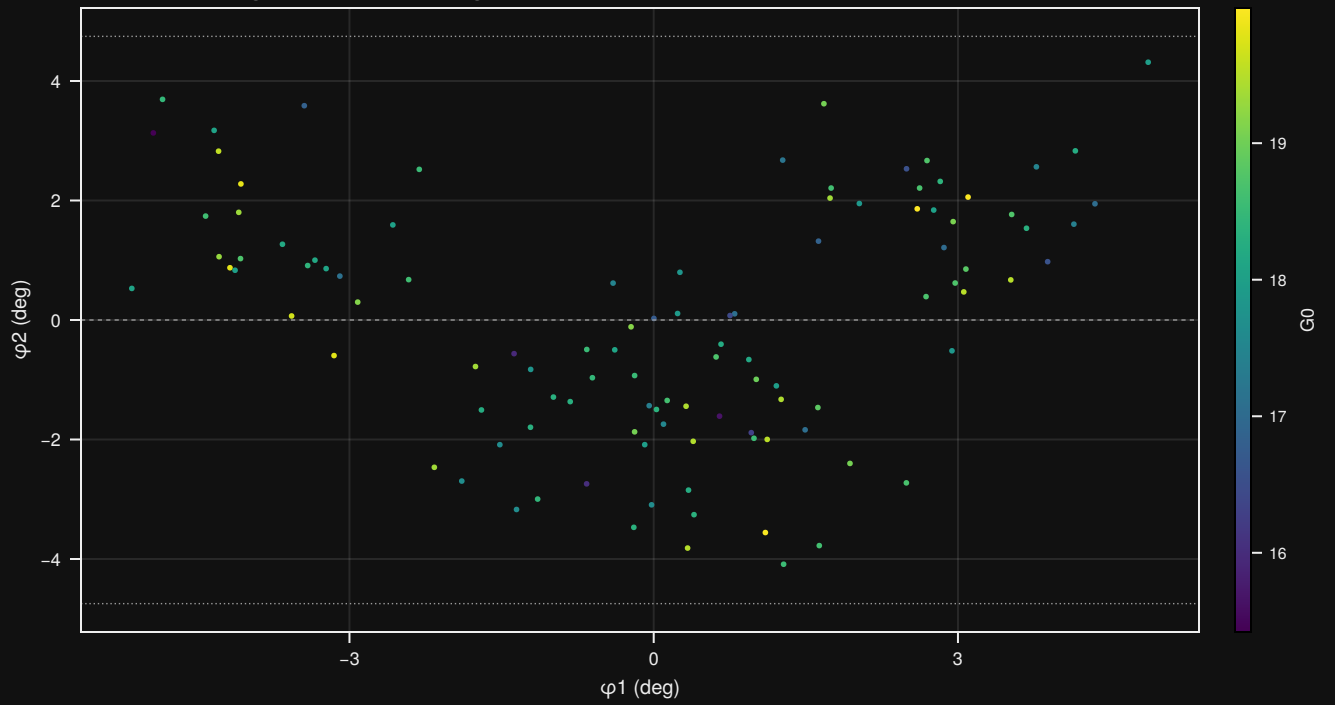
New-18

MEMBERSHIP	
Stars in catalog	107
with Gaia GSP-Phot	79 (74%)
with radial velocity	1
Stream id (sID)	58
GEOMETRY · great-circle fit	
Centroid RA, Dec	213.0, 21.2 deg
Centroid l, b	18.9, 70.0 deg
Pole RA, Dec	122.4, 1.8 deg
Length (φ1, 2–98%)	8.9 deg
Width σ(φ2), robust	2.37 deg
KINEMATICS	
pmRA* med ± σ	-3.42 ± 1.25 mas/yr
pmDE med ± σ	-3.49 ± 2.07 mas/yr
μφ1* med ± σ	-3.63 ± 2.12 mas/yr
μφ2 med ± σ	3.26 ± 1.24 mas/yr
Parallax (cat.) med	0.289 ± 0.188 mas
Parallax (wtd, ZP-corr.)	0.325 ± 0.015 mas
Distance	d = 3.1 ± 0.1 kpc
Physical length	0.48 kpc
RV median ± σ(MAD)	3.9 ± 0.0 km/s
RV gradient dV/dφ1	—
PHOTOMETRY · POPULATIONS	
G0 range (median)	15.4–20.0 (18.3)
(BP−RP)0 median	0.97
GSP-Phot [M/H] (n=79)	-1.17 ± 0.57
GSP-Phot Teff median	5231 K
RUWE > 1.4	0.0 %
PM clusters (largest)	8 (11 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	107 | 107
n_RV catalog | paper	1 | 1
<[Fe/H]> paper	-0.94
Stellar mass (paper)	280 Msun
Pole offset vs paper	0.1 deg

Sky map · catalog members (n = 107) · size ∝ brightness · ring = has RV



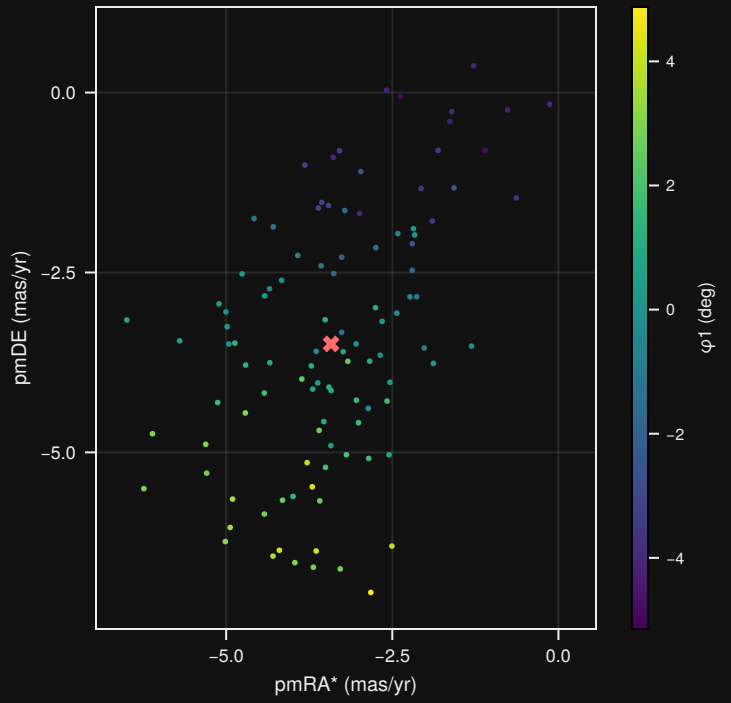
Stream frame (great-circle fit) · length 8.9° · width σ 2.37°



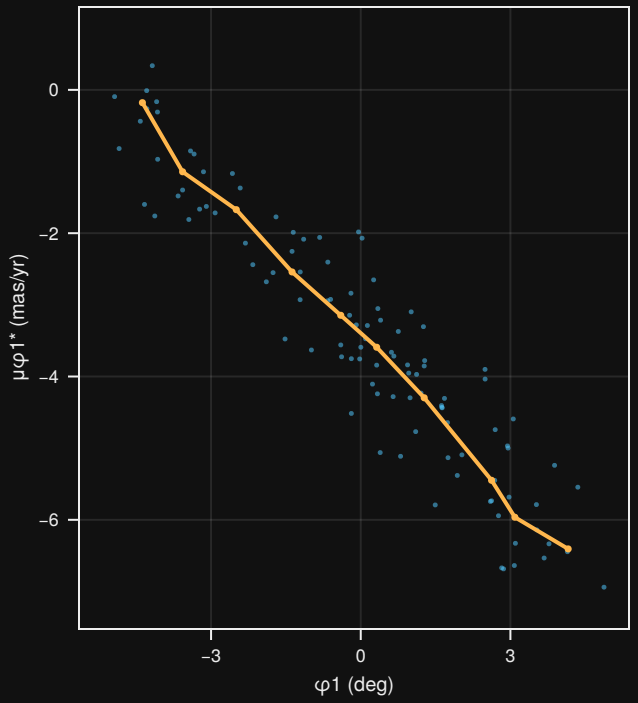
New-18

Kinematics · proper motion · parallax · radial velocity

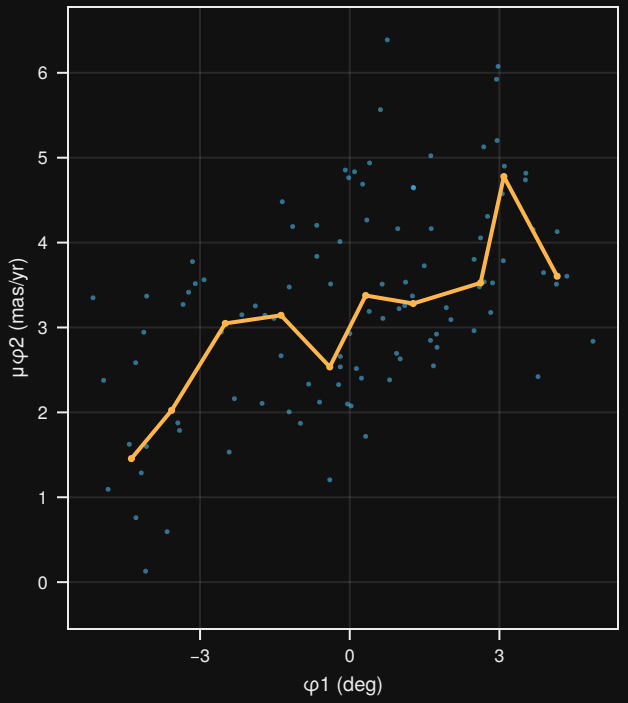
Proper motion (n = 107)



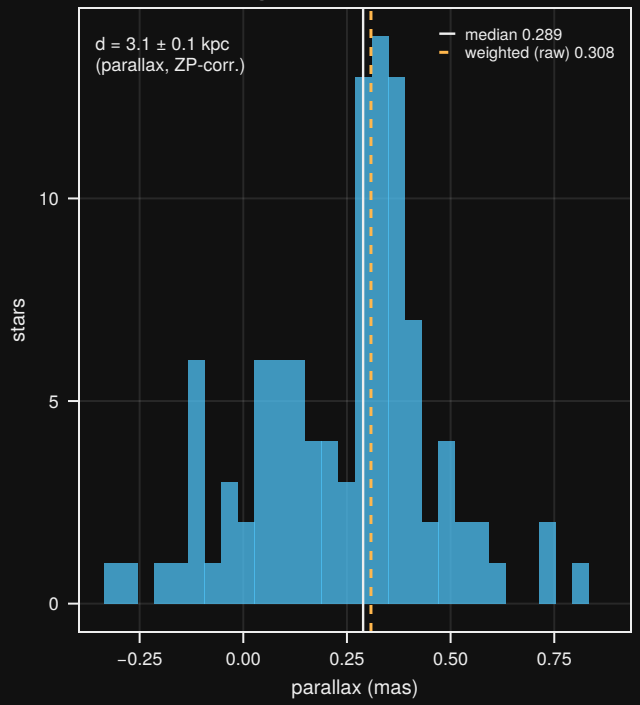
μφ1* vs φ1 · σ = 2.12



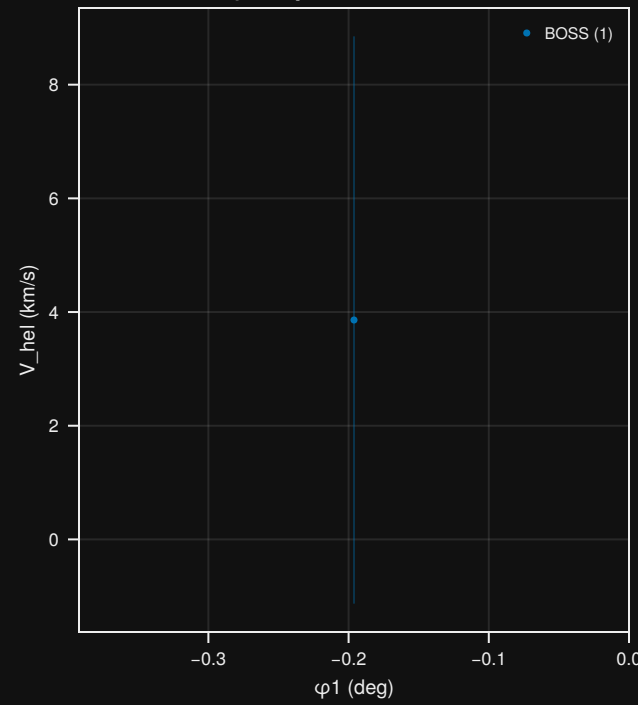
μφ2 vs φ1 · σ = 1.24



Parallax (catalog, n = 107)



Radial velocity vs φ1 (n = 1)



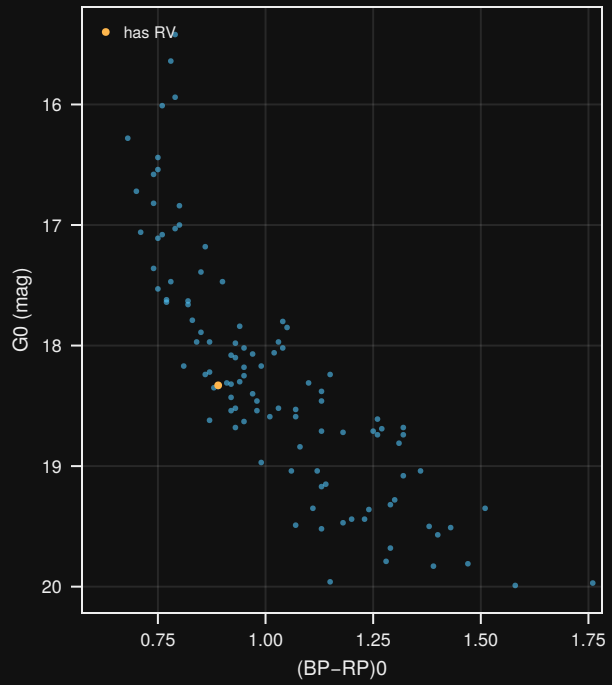
V_hel distribution

n_RV = 1 (< 3)

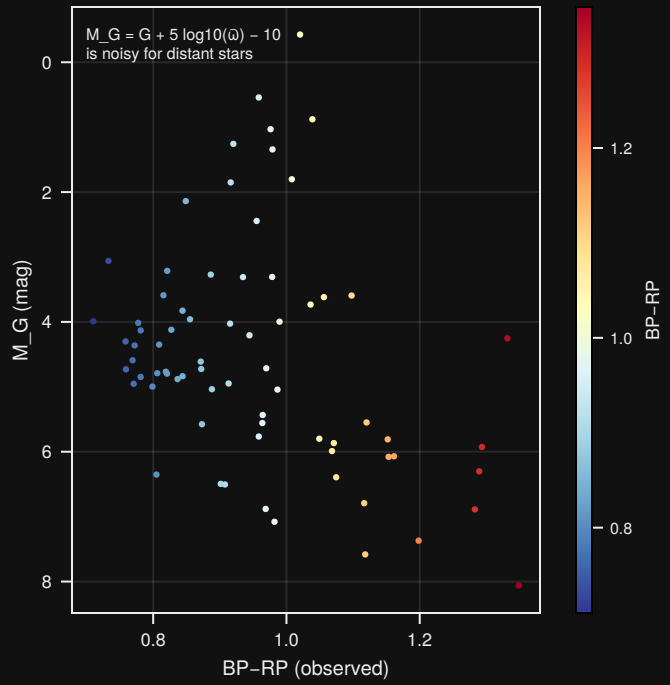
New-18

Photometry · stellar types

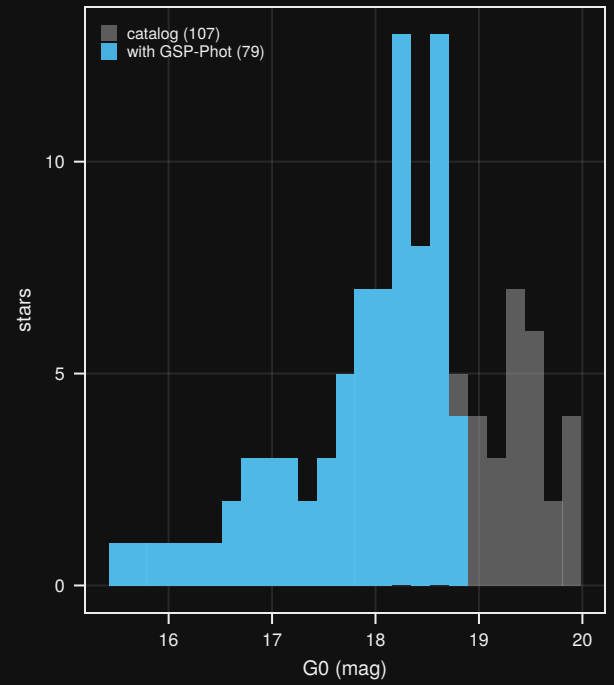
CMD · dereddened (n = 107)



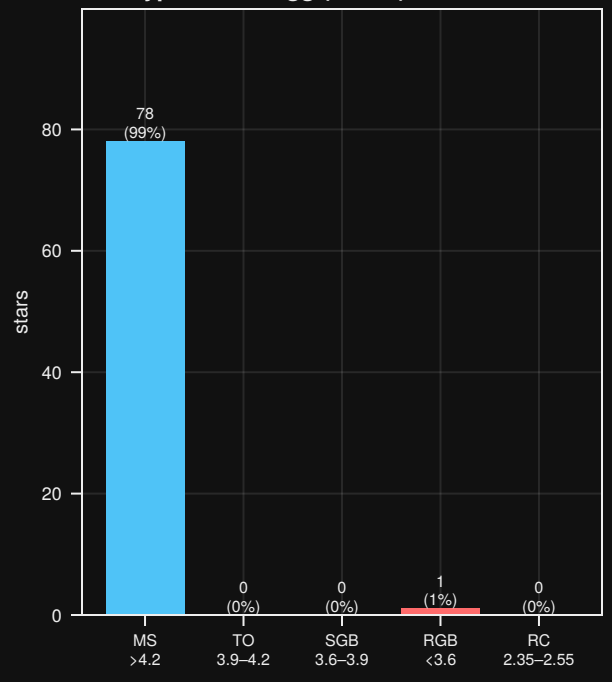
HR diagram · ϖ>0 (n = 72/79)



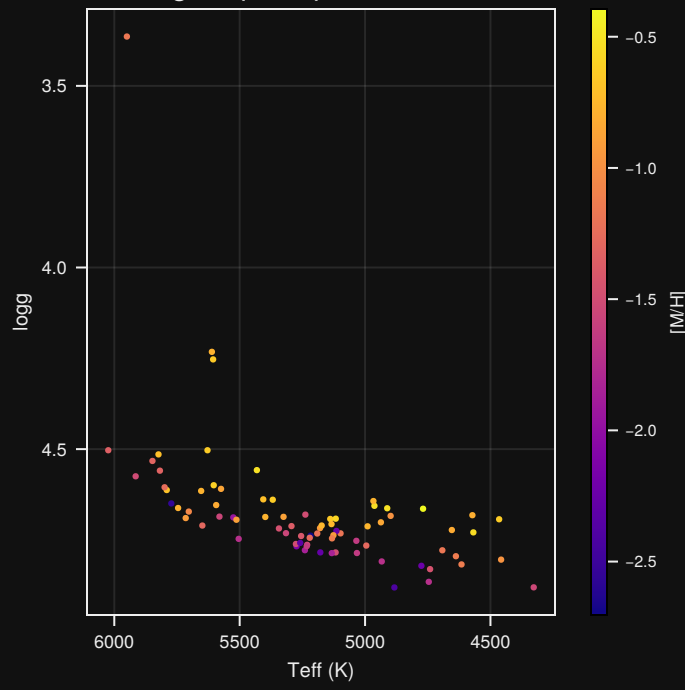
G0 & GSP-Phot coverage



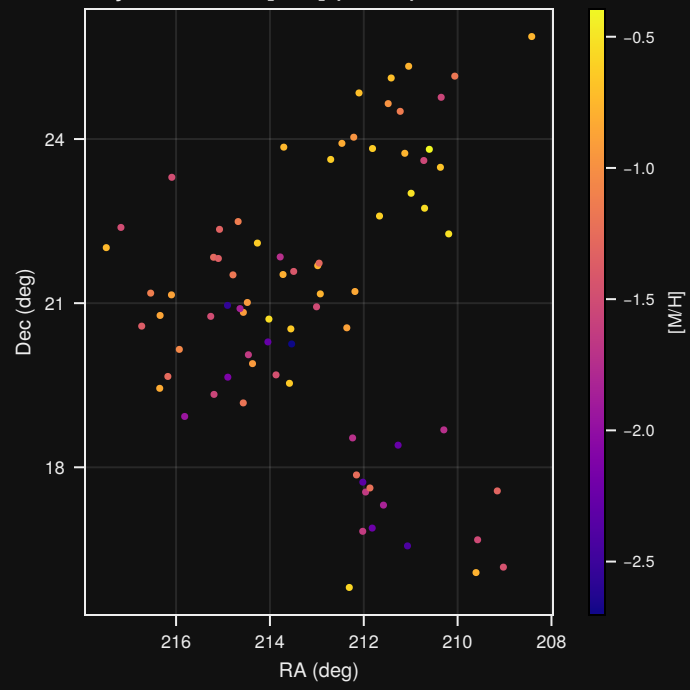
Star types from logg (n = 79)



Kiel diagram (n = 79)



Sky · GSP-Phot [M/H] (n = 79)

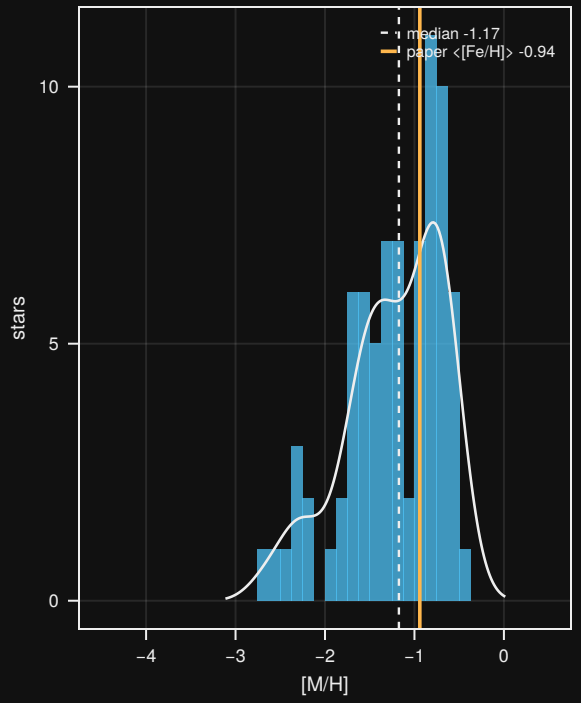


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

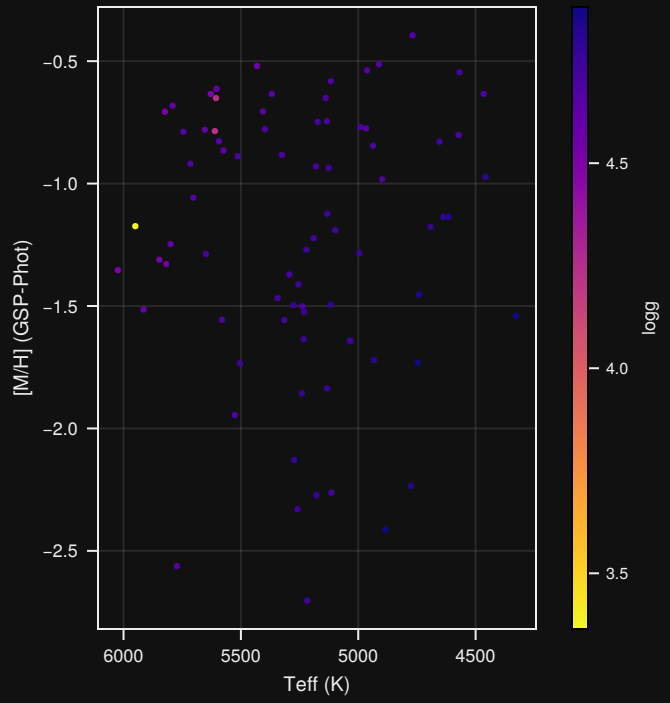
New-18

Metallicity · age · astrometric quality

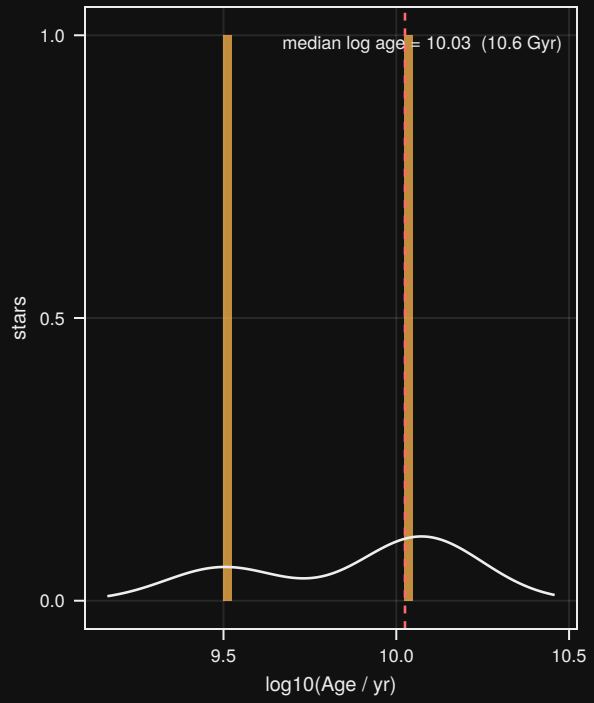
[M/H] GSP-Phot (n = 79)



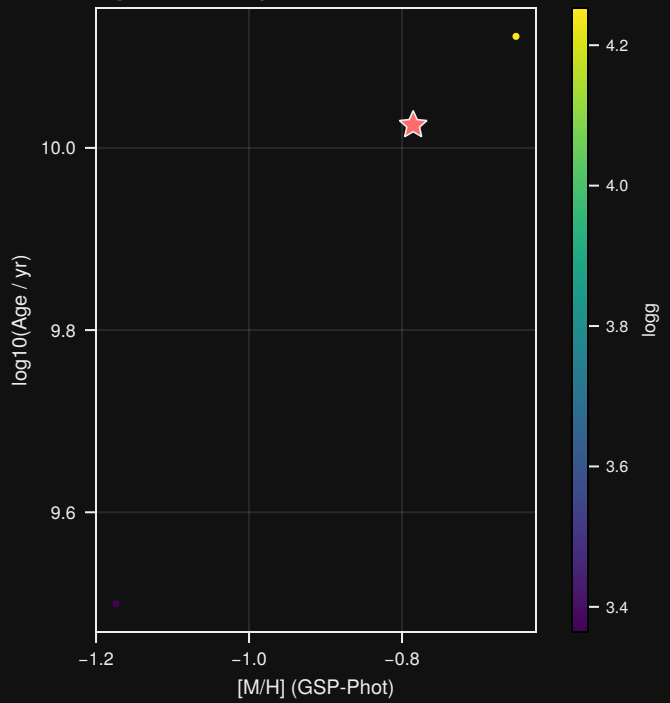
Teff vs [M/H] (n = 79)



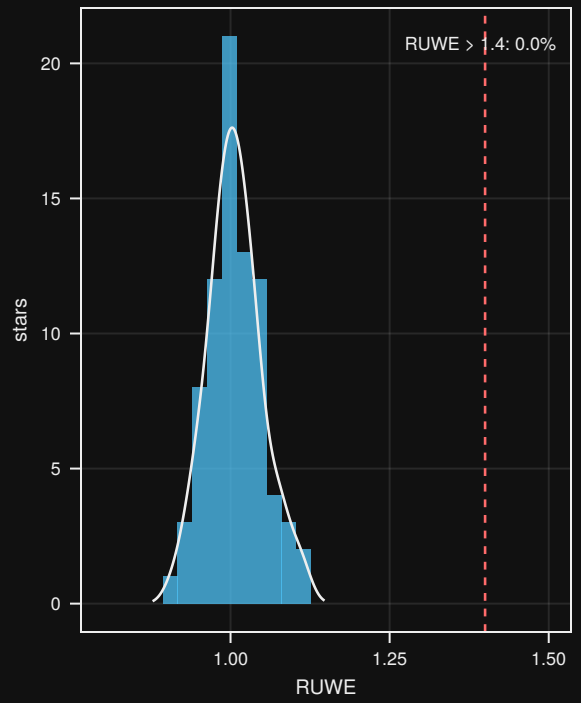
FLAME age (n = 3)



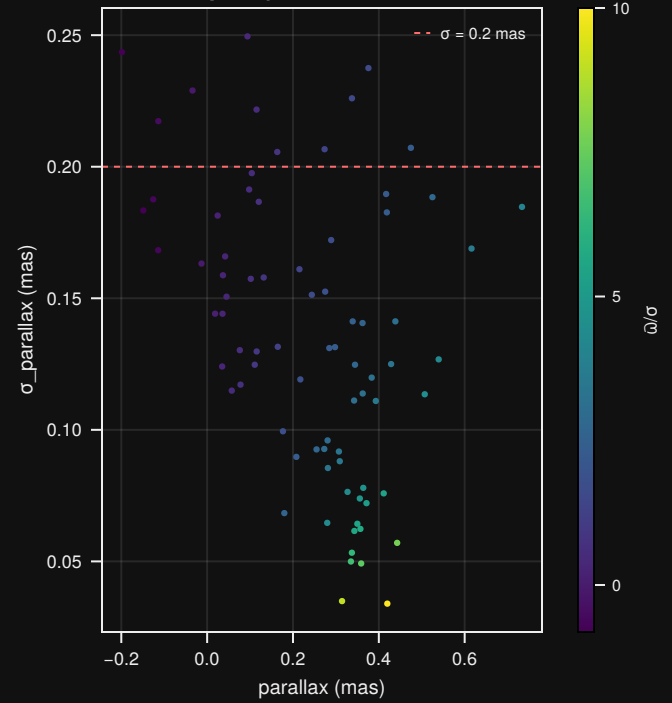
Age–metallicity (n = 3)



RUWE (n = 79)

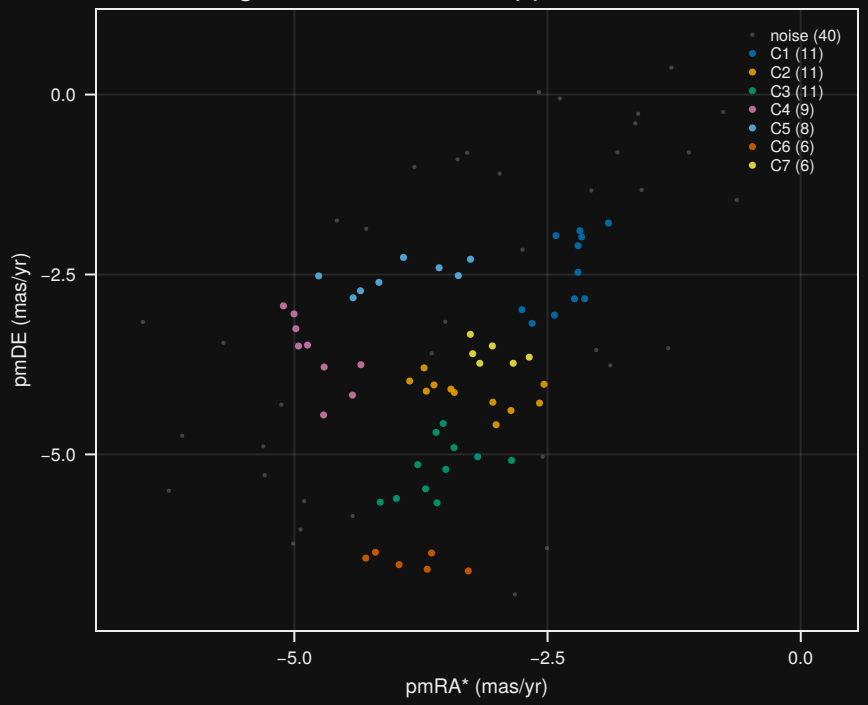


Parallax quality (n = 79)

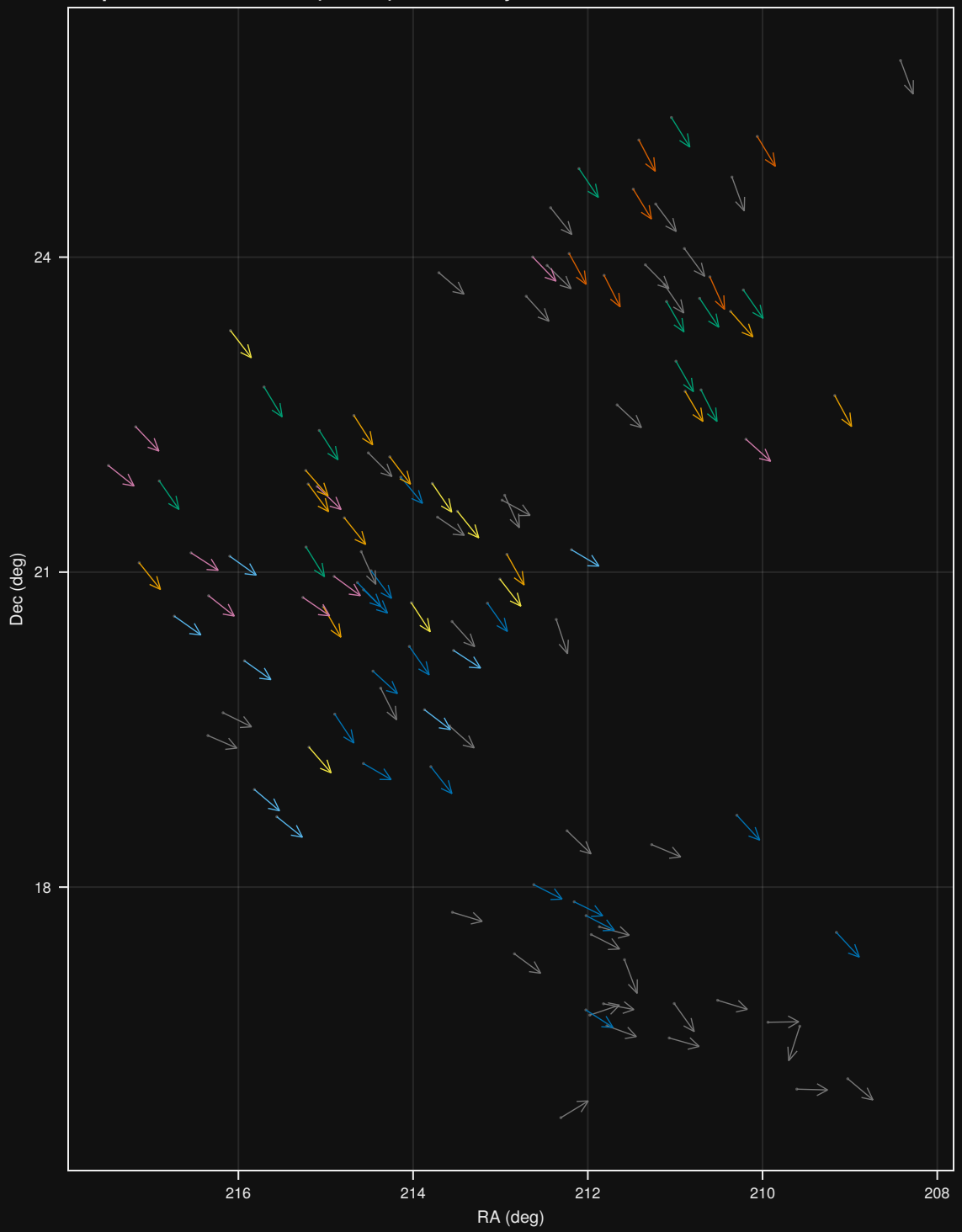


New-18

PM clustering · HDBSCAN · 8 cluster(s) · noise 37%



Proper-motion directions (arrows) coloured by cluster



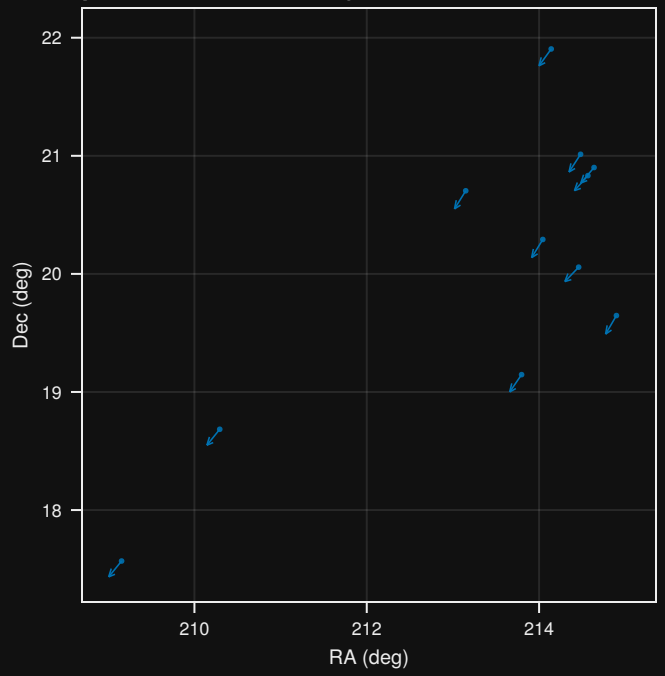
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	40	-2.90	-2.01	2.02	-0.91	0
C1	11	-2.20	-2.47	0.10	-1.69	0
C2	11	-3.42	-4.12	0.56	-0.96	0
C3	11	-3.59	-5.14	0.25	-0.74	0
C4	9	-4.87	-3.49	0.24	-1.24	0
C5	8	-4.04	-2.52	0.63	-1.37	1
C6	6	-3.83	-6.49	0.41	-0.84	0
C7	6	-3.11	-3.62	0.21	-1.50	0
C8	5	-3.46	-1.61	0.23	-1.44	0

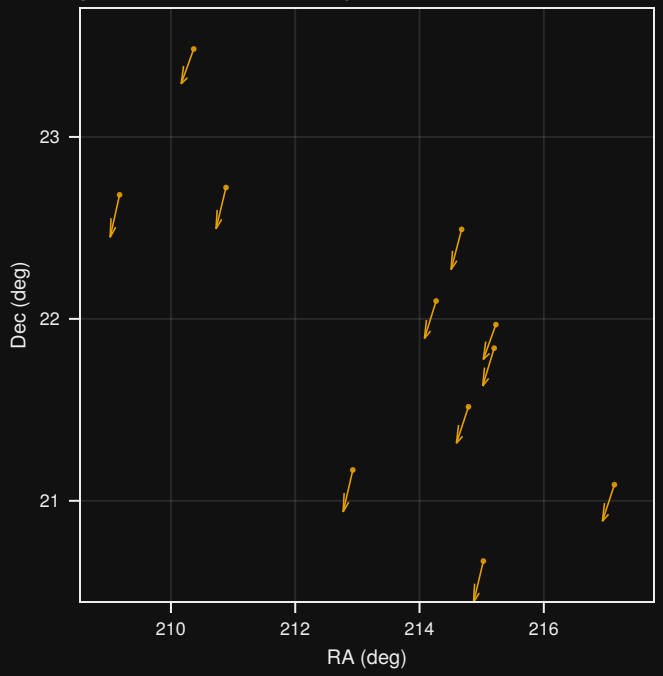
New-18

Per-cluster proper-motion vector maps

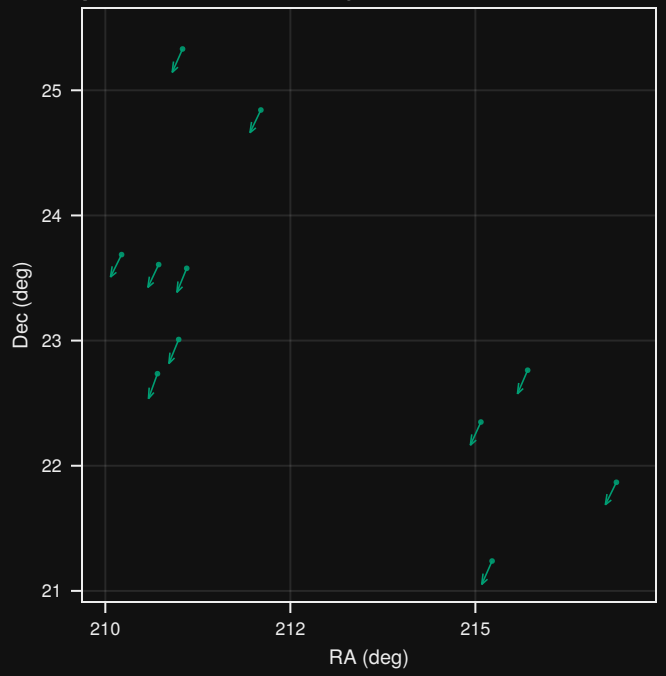
Cluster C1 · n = 11
pm = (-2.20, -2.47) mas/yr



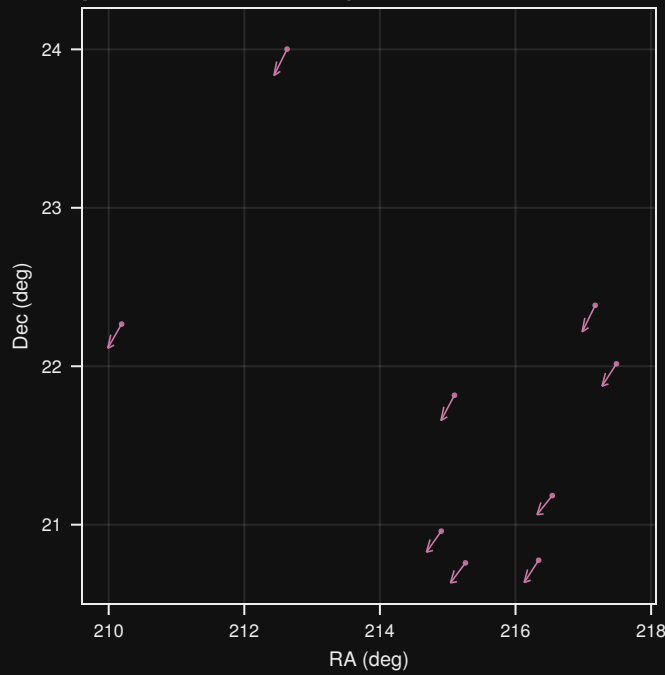
Cluster C2 · n = 11
pm = (-3.42, -4.12) mas/yr



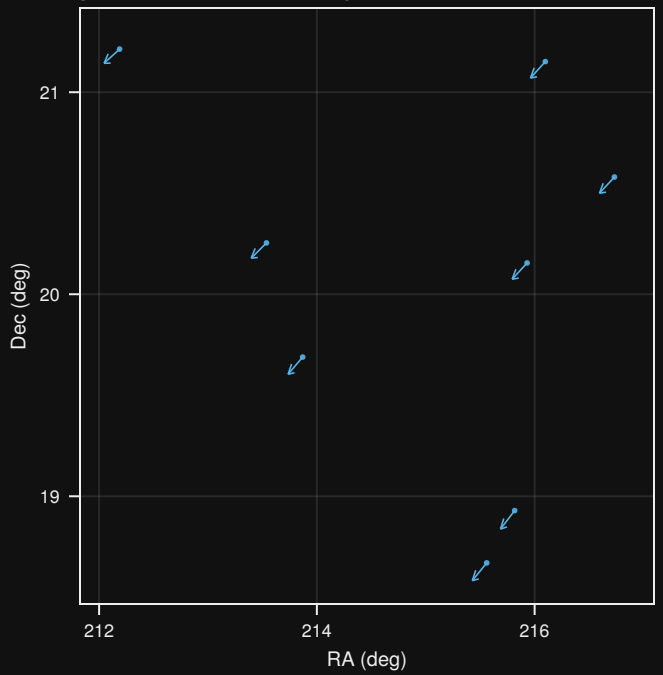
Cluster C3 · n = 11
pm = (-3.59, -5.14) mas/yr



Cluster C4 · n = 9
pm = (-4.87, -3.49) mas/yr



Cluster C5 · n = 8
pm = (-4.04, -2.52) mas/yr



Noise · n = 40

